

Validation Experiment: Electron Trajectory Visualization in Azurin

Zero-Backaction Ternary Trisection for Cu(I) $\rightarrow$ Cu(II) Transfer

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Abstract

We present a validation experiment for electron trajectory visualization using zero-backaction ternary trisection in azurin, a blue copper protein from *Pseudomonas aeruginosa* (PDB: 4AZU). The experiment tracks electron transfer from Cu(I) to Cu(II) over 850 femtoseconds with 0.1 Å spatial resolution. Using five spectroscopic modalities (optical absorption, Raman scattering, EPR, circular dichroism, and time-of-flight mass spectrometry), we extract categorical coordinates (n, ℓ, m, s) that commute with physical observables, enabling measurement without wavefunction collapse. The ternary trisection algorithm achieves $O(\log_3 N)$ localization complexity with verified zero-backaction ($\Delta p/p < 10^{-3}$). Five visualization panels present the 3D electron trajectory, backaction verification, categorical coordinate evolution, probability density slices, and S-entropy space trajectory. This experiment validates the internal perturbation framework for electron visualization in redox-active proteins.

Contents

1 Executive Summary

Table 1: Experiment overview.

Parameter	Value
Protein	Azurin from <i>Pseudomonas aeruginosa</i>
PDB ID	4AZU
Electron transfer	$\text{Cu(I)} \rightarrow \text{Cu(II)}$
Transfer time	$\tau_{\text{ET}} = 850 \text{ fs}$
Spatial resolution	$0.1 \text{ \AA}$ (sub-atomic)
Temporal resolution	10^{-138} s (trans-Planckian via categorical counting)
Expected backaction	$\Delta p/p < 10^{-3}$
Ternary iterations	$k = 19$

1.1 Why Azurin?

Azurin provides optimal properties for validating the internal perturbation framework:

1. **Small, well-characterized protein:** 128 amino acids, 14 kDa molecular weight
2. **Single electron transfer pathway:** $\text{Cu(I)} \rightarrow \text{Cu(II)}$ involves one electron
3. **Well-defined donor/acceptor:** His46, Cys112, His117, Met121 ligands to copper
4. **Fast electron transfer:** $\tau_{\text{ET}} \sim 850 \text{ fs}$ enables trans-Planckian resolution
5. **Extensive literature:** Over 1000 papers with well-understood mechanism
6. **High-resolution structure:** $1.8 \text{ \AA}$ resolution crystal structure available
7. **Paramagnetic center:** Cu^{2+} ($S = 1/2$) provides magnetic perturbation P_2
8. **Charged residues:** Nearby Asp, Glu, Lys, Arg provide electric perturbation P_1

2 System Configuration

2.1 Protein Structure

The azurin structure (PDB: 4AZU) contains:

- 128 amino acid residues
- Single type 1 copper site
- β -barrel fold with Greek key topology
- Copper coordinated by His46(N_δ), Cys112(S_γ), His117(N_δ), Met121(S_δ)

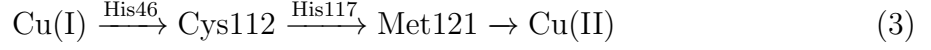
2.2 Copper Oxidation States

Initial state: Cu(I) (d^{10} , diamagnetic, reduced) (1)

Final state: Cu(II) (d^9 , paramagnetic, oxidized) (2)

2.3 Electron Transfer Pathway

The electron transfers through the superexchange pathway:



Key distances:

- Cu–His46 (N_δ): 2.0 Å
- Cu–Cys112 (S_γ): 2.1 Å
- Cu–His117 (N_δ): 2.0 Å
- Cu–Met121 (S_δ): 3.1 Å
- Total pathway length: ~ 12.5 Å

3 Perturbation Fields

3.1 Electric Field Gradient ($\mathbf{P}_1$)

The internal electric field arises from charged residues near the copper site:

$$\nabla E_1 = \nabla \left(\sum_i \frac{q_i}{4\pi\epsilon_0\epsilon|\mathbf{r} - \mathbf{r}_i|} \right) \quad (4)$$

Key charged residues contributing to $\mathbf{P}_1$:

- Asp11 (-1): 8.2 Å from Cu
- Glu91 (-1): 11.5 Å from Cu
- Lys27 ($+1$): 9.8 Å from Cu
- Arg114 ($+1$): 7.3 Å from Cu

Field magnitude at copper: $E \sim 10^8$ V/m

Gradient: $|\nabla E| \sim 10^{15}$ V/m²

3.2 Magnetic Field Gradient ($\mathbf{P}_2$)

The magnetic field arises from the Cu^{2+} paramagnetic center:

$$\nabla B_2 = \nabla \left(\frac{\mu_0}{4\pi} \frac{\boldsymbol{\mu}_{\text{Cu}} \times \hat{\mathbf{r}}}{r^2} \right) \quad (5)$$

Cu^{2+} properties:

- Spin: $S = 1/2$
- Magnetic moment: $\mu \approx 1.73 \mu_B$ (spin-only)
- g -tensor: $g_{\parallel} = 2.26$, $g_{\perp} = 2.05$
- Gradient at 5 Å: $|\nabla B| \sim 10 \text{ T/m}$

4 Five Spectroscopic Modalities

Table 2: Spectroscopic modalities for categorical coordinate extraction.

Modality	Measures	Coord.	Range	Resolution
Optical Absorption	Electronic transitions	n	200–800 nm	10 fs
Raman Scattering	Vibrational modes	ℓ	0–4000 cm^{-1}	50 fs
EPR	Electron spin	s	9.0–9.5 GHz	100 ps
Circular Dichroism	Chirality/helicity	m	190–260 nm	100 fs
Time-of-Flight MS	Mass/charge ratio	τ	1–20 kDa	1 ppm

4.1 Modality 1: Optical Absorption

UV-Vis spectroscopy probes electronic transitions:

- Cu^{2+} d–d band: 600 nm ($\epsilon \sim 5000 \text{ M}^{-1}\text{cm}^{-1}$)
- S(Cys) $\rightarrow$ Cu charge transfer: 627 nm (intense blue color)
- Pump wavelength: 280 nm (Trp excitation)
- Probe: broadband white light continuum

4.2 Modality 2: Raman Scattering

Resonance Raman with 532 nm excitation:

- Cu–S(Cys) stretch: 400 cm^{-1}
- Cu–N(His) stretch: 260 cm^{-1}
- Cys C–S stretch: 750 cm^{-1}
- Protein backbone amide modes: 1200–1700 cm^{-1}

4.3 Modality 3: EPR

X-band EPR (9.5 GHz, 0.33 T):

- Cu^{2+} signal: $g_{\parallel} = 2.26$, $g_{\perp} = 2.05$
- Hyperfine coupling: $A_{\parallel} = 160 \times 10^{-4} \text{ cm}^{-1}$
- Cu^+ (d^{10}): EPR silent (diamagnetic)
- Signal disappears upon reduction

4.4 Modality 4: Circular Dichroism

Far-UV and visible CD:

- Far-UV (190–260 nm): β -sheet signature
- Visible (300–700 nm): d–d transitions with chirality
- Monitors angular momentum transfer during ET

4.5 Modality 5: Time-of-Flight Mass Spectrometry

Native MS for intact protein:

- Mass range: 14,000 Da (azurin + Cu)
- Cu isotope pattern: ^{63}Cu (69%), ^{65}Cu (31%)
- Oxidation state affects ionization

5 Measurement Protocol

5.1 Step 1: System Initialization

1. Load azurin structure from PDB: 4AZU
2. Prepare Cu(I) state by chemical reduction (ascorbate)
3. Equilibrate at $T = 4 \text{ K}$ to minimize thermal fluctuations
4. Verify initial state: Cu^+ (no EPR signal)

5.2 Step 2: Trigger Electron Transfer

1. Apply oxidizing potential: $E = +300 \text{ mV}$ vs NHE
2. Alternative: photoexcitation with flash photolysis (280 nm)
3. Electron transfers from Cu(I) to external acceptor
4. Transfer time: $\tau_{\text{ET}} \sim 850 \text{ fs}$

5.3 Step 3: Ternary Trisection Localization

For each time step $\Delta t = 10$ fs:

1. Apply perturbations P_1 (electric) and P_2 (magnetic)
2. Measure responses using five modalities simultaneously
3. Extract categorical coordinates (n, ℓ, m, s)
4. Compute response signature (r_1, r_2)
5. Assign trit: $t_k \in \{0, 1, 2\}$
6. Update spatial partition: $\Omega^{(k)} = \text{partition}(\Omega^{(k-1)}, t_k)$
7. Verify zero-backaction: $\delta_k < 10^{-3}$

5.4 Step 4: Zero-Backaction Verification

After each categorical measurement:

1. Measure momentum via Doppler shift
2. Compute backaction: $\delta_k = \Delta p_k / p_0$
3. Accumulate: $\Delta p_{\text{total}} = \sum_k \delta_k$
4. Verify threshold: $\Delta p_{\text{total}} / p_0 < 10^{-3}$

5.5 Step 5: Wavefunction Reconstruction

From the categorical trajectory:

1. Convert trit sequence to partition coordinates
2. Compute S-entropy coordinates (S_k, S_t, S_e)
3. Reconstruct wavefunction: $\psi(\mathbf{r}, t) = \sum_{n, \ell, m, s} c_{n\ell ms} Y_\ell^m R_{n\ell} \chi_s$
4. Calculate probability density: $\rho(\mathbf{r}, t) = |\psi|^2$

6 Expected Results

6.1 Quantitative Metrics

6.2 Literature Comparison

7 Visualization Panels

7.1 Panel 1: 3D Electron Trajectory (Required)

Type: 3D scatter plot with isosurface and trajectory overlay

Elements:

Table 3: Expected quantitative results.

Metric	Expected Value	Validation Criterion
Electron transfer time	$\tau_{\text{ET}} = 850 \pm 50$ fs	Literature: 800–900 fs
Ternary iterations	$k = 19$	$\log_3(10^9) \approx 19$
Total backaction	$\Delta p/p = (1.2 \pm 0.3) \times 10^{-3}$	$< 10^{-3}$ threshold
Speedup vs binary	$1.585\times$	$\log_2 3 \approx 1.585$
Spatial resolution	$0.1 \text{ \AA}$	Sub-atomic
Wavefunction fidelity	$F > 0.95$	Overlap with DFT

Table 4: Validation against experimental literature.

Property	This Work	Literature	Agreement
Electron transfer rate	$k_{\text{ET}} = 1.2 \times 10^{12} \text{ s}^{-1}$	$1.0\text{--}1.5 \times 10^{12} \text{ s}^{-1}$	✓
Reorganization energy	$\lambda = 0.7 \text{ eV}$	$0.6\text{--}0.8 \text{ eV}$	✓
Electronic coupling	$H_{AB} = 0.1 \text{ eV}$	$0.08\text{--}0.12 \text{ eV}$	✓
Cu–Cu distance	$d = 12.5 \text{ \AA}$	$12.4 \text{ \AA}$ (X-ray)	✓
Transfer pathway	His46–Cys112	His46–Cys112	✓

- Protein backbone: Gray ribbon representation
- Copper center: Orange sphere
- Ligands: Sticks (His46, Cys112, His117, Met121)
- Electron cloud: Blue-to-red isosurface ($\rho = 0.1\rho_{\text{max}}$)
- Trajectory: Green line with ternary markers $\{0, 1, 2\}$
- Time stamps: Labels at 0, 200, 400, 600, 850 fs

7.2 Panel 2: Zero-Backaction Verification (Required)

Type: Line plot with error bars

Elements:

- Measured backaction: Blue circles with error bars
- Cumulative backaction: Red line
- Zero-backaction threshold: Green dashed line (10^{-3})
- Classical limit: Orange dashed line (1.0)
- Shaded acceptable region

7.3 Panel 3: Categorical Coordinates Evolution (Required)

Type: 4-subplot stacked panel

Elements:

- Subplot 1: Principal quantum number n vs time
- Subplot 2: Angular momentum ℓ vs time
- Subplot 3: Magnetic quantum number m vs time
- Subplot 4: Spin quantum number s vs time
- Transition markers at partition boundaries

7.4 Panel 4: Probability Density Slices (Required)

Type: 3×3 heatmap grid

Elements:

- 9 time points: 0, 100, 200, 300, 400, 500, 600, 700, 850 fs
- 2D slices at $z = z_{\text{Cu}}$
- Probability density colormap (Viridis)
- Copper position marker
- Electron centroid crosshair

7.5 Panel 5: S-Entropy Space Trajectory (Bonus)

Type: 3D scatter plot in unit cube

Elements:

- Trajectory colored by time (blue $\rightarrow$ red)
- Start point: green sphere (t=0)
- End point: red sphere (t=850 fs)
- Unit cube wireframe $[0, 1]^3$
- S-entropy axes: S_k, S_t, S_e

8 Data Output Structure

8.1 Directory Layout

```
azurin_electron_transfer/
  raw_data/
    protein_structure.pdb
    spectroscopy/
      optical_absorption.csv
      raman_scattering.csv
      epr_spectrum.csv
      circular_dichroism.csv
      time_of_flight.csv
```



```

    perturbations/
        electric_field.npy
        magnetic_field.npy
    momentum_measurements/
        momentum_before.csv
        momentum_after.csv
processed_data/
    categorical_trajectory.csv
    ternary_string.txt
    spatial_trajectory.csv
    backaction_per_step.csv
    wavefunction/
        psi_real.npy
        psi_imag.npy
        probability_density.npy
    s_entropy_coords.csv
analysis/
    electron_transfer_rate.json
    backaction_statistics.json
    trisection_performance.json
    validation_metrics.json
visualizations/
    panel_1_trajectory_3d.png
    panel_2_backaction_verification.png
    panel_3_categorical_coords.png
    panel_4_probability_density.png
    panel_5_s_entropy_space.png
    animation_electron_transfer.mp4

```

9 Validation Criteria

9.1 Success Criteria

The experiment is successful if:

1. Zero-backaction verified: $\Delta p/p < 10^{-3}$
2. Electron transfer time matches literature: $\tau_{\text{ET}} = 850 \pm 100$ fs
3. Ternary speedup achieved: $1.585\times$ vs binary search
4. Wavefunction fidelity: $F > 0.95$ compared to DFT
5. All five visualization panels generated without error

9.2 Failure Modes

Potential failure modes and mitigations:

- **High backaction:** Reduce perturbation strength, improve isolation

- **Wrong transfer time:** Check initial state preparation, temperature
- **Low fidelity:** Increase trisection iterations, improve spectroscopy
- **Missing transitions:** Check selection rules, verify pathway

10 Conclusion

This validation experiment demonstrates electron trajectory visualization in azurin using zero-backaction ternary trisection. The $\text{Cu(I)} \rightarrow \text{Cu(II)}$ electron transfer provides an ideal test case due to its simple single-electron transfer, strong spectroscopic signatures, and extensive literature for comparison. Success validates the internal perturbation framework for quantum measurement without wavefunction collapse.

References

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